Time Frequency Processing Python Project

Project and description generated with the help of ChatGPT.

One week project for senior and first year graduate electrical and computer engineering students. Work individually. Budget approximately 12 to 15 hours to complete. Use NumPy, SciPy and Matplotlib; implement the three core DSP functions yourself. The supplied waveform generator and matched QPSK receiver keep the work focused on DSP rather than audio collection or modem design.

Your task is to analyze time varying signals, remove a jammer, and reconstruct signals without losing phase coherence. Explain how window duration, overlap and spectral leakage affect the result. Connect your reasoning to Oppenheim and Schafer topics on the DFT, windowing, FFT convolution and short time Fourier analysis. Section numbers depend on the textbook edition.

Expected preparation

You should know complex DFT coefficients, convolution, FIR filtering and the sampling theorem. Read the relevant textbook sections before starting. No machine learning, blind synchronization or real time hardware is required.

Day	Milestone	Hours
1	Inspect signals and implement STFT analysis	2
2	Window and uncertainty experiments	2.5
3	Reassembly and overlap tests	2
4	FFT overlap add and jammer removal	3.5
5 to 7	QPSK tests (ECEN562) and report	3

Submission

Submit one executed notebook, your student_dsp.py, and a 5 to 7 page report with labeled graphs. Include requirements and a fixed random seed. Save constellation plots and numerical metrics. Acknowledge any additional resources that you used, including AI (AI company, text or description of prompt used). Do not submit an entirely AI generated solution.

Use a common amplitude scale for audio comparisons, label axes and units, and state sample rate, window, frame length, hop and FFT length on every time frequency figure. Label magnitude displays as relative dB; do not call them calibrated PSD estimates unless you implement the proper normalization.

Signals and Transform Conventions

All supplied test WAV files use 16000 samples per second. Float32 WAV files are deliberate; modern scientific readers support them. Use common.read_wav to handle ordinary integer PCM files and sample rate conversion. Compute in float64 or complex128.

File	Purpose
tf_probe.wav	Close tones at 1000 and 1080 Hz, a chirp and a brief transient
leakage.wav	1024 samples with a strong 1003 Hz off bin tone and a weak 1140 Hz tone
melody.wav	Desired synthetic plucked note sequence with attacks and harmonics
jammed.wav	Melody plus a chirp sweeping from 1800 toward 3000 Hz over four seconds
jammer.wav	Evaluation only; never use to estimate the ridge
qpsk_iq.wav	Stereo complex baseband: left channel I and right channel Q

signals/truth.npz provides evaluation references and modem metadata. You may use clean signals to calculate scores.

Local index STFT definition

$$X_m[k] = \sum_{r=0}^{L-1} x[mH + r]w[r]e^{-j2\pi kr/N}$$

L is frame length, H is hop and N is FFT length with N at least L. Overlap fraction is 1 minus H divided by L. Use a periodic analysis window. Store complex coefficients, original signal length and padding. Frame times use the window center; subtract the initial padding offset. For I/Q use a two sided FFT and retain complex samples.

Part A Time Frequency Analysis and Windows

A1 Window duration and uncertainty 15 points

Implement analysis. On tf_probe.wav compare Hann windows of 256, 1024 and 4096 samples, each with 75 percent overlap and FFT length equal to L. Plot three spectrograms with identical frequency limits and dB ranges. Report frame duration, bin spacing and hop time. Inspect the two close tones away from transients and quantify the observed transient width using an explicitly stated threshold or energy criterion.

$$T_w = L/f_s, \qquad \Delta f_{\text{bin}} = f_s/N, \qquad T_H = H/f_s$$

Explain why long windows help distinguish close steady tones but smear short events and broaden the apparent sweep of a chirp within each frame. FFT bin spacing is not the true resolving power. At L=1024, repeat with N=4096 while keeping L and H fixed. Explain what zero padding changes and what it cannot change. Use the time frequency uncertainty idea qualitatively; a continuous time Gaussian bound should not be applied blindly to finite periodic DFT samples.

A2 Spectral leakage 15 points

On leakage.wav compare rectangular, periodic Hann, Hamming and Blackman windows with L=1024 and N=65536. Divide amplitudes by the window sum so comparisons account for coherent gain. Plot amplitudes in dB relative to 1. Explain whether the weak neighboring tone is masked by leakage.

Also compute each window transform alone, normalized by its DC amplitude. Measure null to null main lobe width and highest sidelobe outside the main lobe. State your numerical definition of a first null or minimum. Calculate equivalent noise bandwidth from the equation below. Do not measure sidelobe levels from the two tone mixture.

$$\text{ENBW} = f_s \frac{\sum_r w^2[r]}{\left(\sum_r w[r]\right)^2}$$

Discuss the tradeoff: lower sidelobes help detect weak signals near strong off bin components, while a wider main lobe reduces separation of nearby components. A window is not universally best. Optionally add a brief natural speech or music example and justify a window duration for that signal.

Part B Reconstruction and Overlap Add Filtering

B1 STFT reassembly 15 points

Implement synthesis with IFFT, a synthesis window, overlap summation, samplewise normalization and exact cropping. For analysis and synthesis using the same real window, use the squared window denominator. Record the minimum and maximum denominator over retained samples.

$$\hat{x}[n] = \frac{\sum_m w[n - mH]\, v_m[n - mH]}{\sum_m w^2[n - mH]}$$

Here v is the first L samples of the N point IFFT of a frame. An unmodified frame equals the analysis window times the input. A nonzero denominator on every retained sample is the NOLA (nonzero overlap add) requirement. Include padding before and after the signal. With modified spectra, this normalized synthesis is a reconstruction operation and generally is not exact LTI convolution.

Compare Hann overlaps of 0, 25, 50, 75 and 87.5 percent at L=1024. Expect zero overlap to fail because the Hann window has zeros; detect and explain the failure instead of dividing by zero. Show a rectangular zero overlap control. For valid cases report relative reconstruction error, frame count and measured runtime. Runtime is descriptive, not a grading threshold. Float64 identity error should be below 1e-10. Report audio WAV round trip error separately.

B2 Frequency domain FIR filtering 15 points

Implement ola_filter using nonoverlapping input blocks of B=512 samples and a 129 tap Hamming lowpass FIR with cutoff 1200 Hz. Zero pad each block, multiply its FFT by the filter FFT, inverse transform, then overlap add the full block convolution outputs. Use the supplied filter design call or an equivalent one; you are not being graded on designing an optimal lowpass.

$$N_{\text{FFT}} \geq B + M - 1, \qquad N_y = N_x + M - 1$$

Filter the “mixture” signal (jammed.wav) with the lowpass FIR and listen to the original and the filtered signals. Verify the full output against np.convolve, including the final partial block and filter tail. Relative error should be below 1e-10. Explain why too short an FFT creates circular aliasing. Repeat for B=128 and 2048 and compare FFT sizes and frame counts. Account for the 64 sample FIR group delay when making waveform comparisons. Do not apply Hann windows or squared window normalization to this exact block convolution method.

Part C Phase Coherence (for ECEN 5632)

C2 Phase coherent communications 15 points

Use supplied 1000 symbol per second QPSK, 16 samples per symbol, root raised cosine pulses with rolloff 0.35 and a 10 symbol span. Reassemble the full I/Q signal with the unchanged STFT, then pass it to the supplied matched receiver. Compare waveform error, EVM, BER and constellation plots for the original and reconstructed signals.

Repeat with two deliberately broken methods: replace every STFT coefficient by its magnitude, and multiply each frame by an independently chosen random phase. No per frame gain normalization or phase correction is allowed. Keep the known timing and supplied matched filter. Explain why an attractive spectrogram does not demonstrate successful QPSK reconstruction. Coherent BER should be zero and EVM should match the original baseline; finite pulse truncation can give a small nonzero baseline EVM.

Assessment and Extensions

Area	Points	Evidence for full credit
Time frequency tradeoff	15	Correct STFT 5; controlled plots and measurements 6; uncertainty and zero padding reasoning 4
Windows and leakage	15	Correct gain and window metrics 6; weak tone experiment 5; justified tradeoffs 4
Overlap reconstruction	15	Padding and normalization 7; overlap sweep and failure control 5; error and cost reasoning 3
FFT overlap add filtering	15	Linear convolution algorithm and tail 8; reference checks 4; aliasing and delay explanation 3
QPSK phase coherence	15	Complex I/Q and receiver comparison 6; broken controls and constellations 5; phase explanation 4
Reproducibility and communication	5	Executable notebook and labeled outputs 3; concise report and contributions 2

Grade correctness and reasoning rather than visual polish or the number of figures. Award partial credit for a correct method with a diagnosed coding error. Identity thresholds apply to in memory float64 computations, not quantized WAV exports. A zero BER requirement applies to the supplied noiseless coherent control, not a new noisy channel.

Graduate depth

For graduate students, require a one page deeper analysis within the same grading rubric: derive the squared window normalization. Seniors may present a correct experimental explanation without the full derivation. Grade this distinction within the explanation points already listed, rather than adding workload unrelated to the DSP objectives.

Resources

Use your course edition of Oppenheim and Schafer. A SciPy reference link is in README.md.